SST Triadic-Casimir / PKLSA Blind Falsifier v0.2.0

Framework-aware, three-source-seal, geometry-only scientific diagnostic

Run label: REPORT

1. Overall epistemic verdict

Analytical toy controls and a sourced KnotPlot/PKLSA-lineage trefoil geometry were evaluated. A physical Casimir force, atom-link coupling, quantum vacuum amplitude, and gravity are NOT certified.

Official frozen-framework invocation: OFFICIAL_FRAMEWORK_EXECUTED

Framework detail: Canonical framework code ran through real generator, protocol freeze, FULL and reveal-policy check. PHYSICAL_G3_UNRESOLVED.

2. Three independently sealed datasets

D01_EULER SHA-256 seal: 94f2ec80b770b7a9d3aeba3905019b2c64ddc7fbed7a23517774f69137a8ffb4

D02_CLOCK_LINK SHA-256 seal: 2e48410caab26aa0d082d56cd21580f18bb0889706182023ef37d94c0bf8c7f5

D03_TWO_SURFACE SHA-256 seal: 42ad55ee2e5385e030343a3610c3b578289181f4e14aec9b5e638c7ac9877a22

3. Numeric control gates

D01_G01_EULER_DIV: PASS | observed=0.0 | criterion=1e-12

D01_G02_POISSON: PASS | observed=0.0 | criterion=1e-12

D01_G03_POTENTIAL_NULL: PASS | observed=0 | criterion=1e-12

D01_G04_IRROTATIONAL_CLOCK_VARIATION: PASS | observed=0.0008849583697424501 | criterion=1e-05

D01_G05_PKLSA_SOURCE_INTEGRITY: PASS | observed=300 | criterion=300

D01_G06_PKLSA_FINITE_BIOTSAVART: PASS | observed=0.002346617627425827 | criterion=finite

D02_G01_GAUGE: PASS | observed=5.551115123125783e-17 | criterion=1e-12

D02_G02_PHASE_RATE: PASS | observed=5.551115123125783e-17 | criterion=1e-12

D02_G03_HOLONOMY: PASS | observed=2.0816681711721685e-17 | criterion=1e-12

D02_G04_RESPONSE_POSITIVE: PASS | observed=0.0 | criterion=0.0

D02_G05_SHEET_STABLE: PASS | observed=1.6305756807341665 | criterion=0.0

D02_G06_NO_RESPONSE_CONTROL: PASS | observed=0.0 | criterion=1e-14

D03_G01_ENERGY_STRESS: PASS | observed=1.141916836020315e-09 | criterion=0.015

D03_G02_DECOUPLED_NULL: PASS | observed=0.0 | criterion=1e-13

D03_G03_FINITE_TOY_RESPONSE: PASS | observed={'negative': 28, 'total': 28} | criterion=all toy coupled configurations

Pilot synthetic/numerical checks: 15/15 PASS

4. PKLSA geometry and numerical bridge

Sourced 3_1 trefoil: KnotPlot relaxation sequence, N=300, one upstream family (NOT two independent providers).

Source Drive id: 1iOeQe6-jHcBk_NYNwZe3e-N2DQPwohmI

Raw geometry SHA-256: a60b1a95a00110e3cb32087e7b4328d16e967379ebf97d60117b0ec05692ffb2

Points=300; length=16.376343183; segment relative spread=0.00069244

Regularized midpoint Biot-Savart: Gamma=1, eps=0.12, max 150 vs 300 refinement error=0.002346618

The pressure-Poisson source at sample points is computed from numerical strain and curl of the regularized geometric velocity field, with no wall/stress certification.

5. Model equations

Euler: div(u)=0; omega=curl(u); Laplace(p)=rho*(|omega|^2/2 - ||S||^2).

Conditional clock: N^2=1-|u_eff|^2/c_star^2. PKLSA geometry does not independently determine u_eff for a clock.

Link clock phase: Theta_AB=theta_B-theta_A-sum(path theta_e), conditional on

gauge-covariant atom coupling.
One-body response: chi(i xi)=bT (K+xi^2 M)^(-1)b, presently a toy oscillator, not measured metal.
Two-body finite-T toy: F/A=(T/4pi) integral k log(1-r1*r2*exp(-2*k*a)) dk.

6. Physical qualification blockers

Missing: independent second PKLSA provider, time-dependent Euler evolution, finite-core constitutive law, laboratory wall boundary conditions, atom-link coupling, independent reflection operator, fluctuation normalization and independent physical stress.
Never interpret fitted distance exponent or numeric toy attraction as a predicted electromagnetic Casimir force.

7. Reproduction and PDF availability

On Windows: run_all.cmd ALL; run_all.cmd FRAMEWORK; run_all.cmd PKLSA; run_all.cmd VERIFY. A current dynamic PDF is created even if LaTeX/ReportLab are absent.
The official canonical SST framework v1.0.7 must exist at the Workbench template location, or SST_FALSIFIER_FRAMEWORK_ROOT must point to it. Its source remains unmodified.
All three v0.2.0 dataset seals were reissued after adding PKLSA bytes to D01. The earlier v0.1.0 seal is NOT reused as if it still covered the same data.

8. Provenance

Canon v0.8.36: sections 3.5, 3.10, 7.1, 11.16-11.17, S.2, S.52.
A046 v0.2.0 and v0.3.0: Euler / Poisson controls; geometry is not gravity.
Original KnotPlot Drive id and full source sha256 are included in D01. All fields are dimensionless; no prescribed SST rho_f or Casimir amplitude.